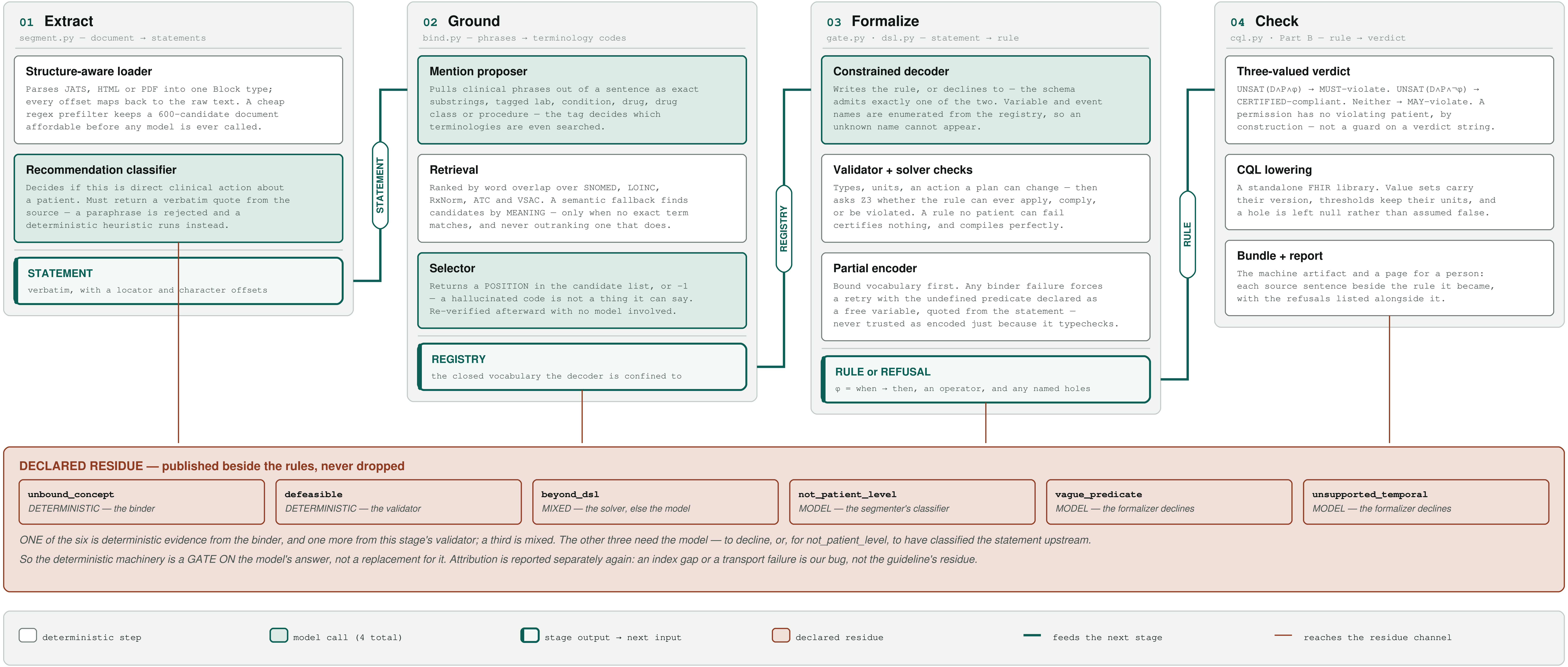

01 Extract

segment.py — document → statements

Structure-aware loader

Parses JATS, HTML or PDF into one Block type; every offset maps back to the raw text. A cheap regex prefilter keeps a 600-candidate document affordable before any model is ever called.

Recommendation classifier

Decides if this is direct clinical action about a patient. Must return a verbatim quote from the source — a paraphrase is rejected and a deterministic heuristic runs instead.

STATEMENT

verbatim, with a locator and character offsets

02 Ground

bind.py — phrases → terminology codes

Mention proposer

Pulls clinical phrases out of a sentence as exact substrings, tagged lab, condition, drug, drug class or procedure — the tag decides which terminologies are even searched.

Retrieval

Ranked by word overlap over SNOMED, LOINC, RxNorm, ATC and VSAC. A semantic fallback finds candidates by MEANING — only when no exact term matches, and never outranking one that does.

Selector

Returns a POSITION in the candidate list, or −1 — a hallucinated code is not a thing it can say. Re-verified afterward with no model involved.

REGISTRY

the closed vocabulary the decoder is confined to

03 Formalize

gate.py · dsl.py — statement → rule

Constrained decoder

Writes the rule, or declines to — the schema admits exactly one of the two. Variable and event names are enumerated from the registry, so an unknown name cannot appear.

Validator + solver checks

Types, units, an action a plan can change — then asks Z3 whether the rule can ever apply, comply, or be violated. A rule no patient can fail certifies nothing, and compiles perfectly.

Partial encoder

Bound vocabulary first. Any binder failure forces a retry with the undefined predicate declared as a free variable, quoted from the statement — never trusted as encoded just because it typechecks.

RULE or REFUSAL

φ = when → then, an operator, and any named holes

04 Check

cql.py · Part B — rule → verdict

Three-valued verdict

$\mathrm{UNSAT}(D \wedge P \wedge \varphi)$ → MUST-violate. $\mathrm{UNSAT}(D \wedge P \wedge \neg\varphi)$ → CERTIFIED-compliant. Neither → MAY-violate. A permission has no violating patient, by construction — not a guard on a verdict string.

CQL lowering

A standalone FHIR library. Value sets carry their version, thresholds keep their units, and a hole is left null rather than assumed false.

Bundle + report

The machine artifact and a page for a person: each source sentence beside the rule it became, with the refusals listed alongside it.

DECLARED RESIDUE — published beside the rules, never dropped

unbound_concept	*DETERMINISTIC — the binder*
defeasible	*DETERMINISTIC — the validator*
beyond_dsl	*MIXED — the solver, else the model*
not_patient_level	*MODEL — the segmenter's classifier*
vague_predicate	*MODEL — the formalizer declines*
unsupported_temporal	*MODEL — the formalizer declines*

ONE of the six is deterministic evidence from the binder, and one more from this stage's validator; a third is mixed. The other three need the model — to decline, or, for not_patient_level, to have classified the statement upstream.

So the deterministic machinery is a GATE ON the model's answer, not a replacement for it. Attribution is reported separately again: an index gap or a transport failure is our bug, not the guideline's residue.

☐	deterministic step
🟩	model call (4 total)
▣	stage output → next input
🟥	declared residue
—	feeds the next stage
—	reaches the residue channel